Saravanan Rajendran

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Professional Summary

Technical Lead Engineer with **7+ years** shipping distributed systems on a **microservices** platform ingesting **600K+ msg/min**; re-architected the primary data pipeline to cut latency **30-60x** at **99.9% SLA**, and built an AI debugging agent with **RAG** over logs/code. Backend in **TypeScript / JavaScript (Node.js)**; prior **Senior SWE @ Myntra**.

Education

University of Illinois Urbana-Champaign — Urbana, IL
Master of Computer Science, AI Track — *August 2026 - June 2028 (Expected)*

Coursework: Distributed Systems, Database Systems, Algorithms, Software Engineering, Machine Learning, Artificial Intelligence.

College of Engineering, Guindy, Anna University — Chennai, India
B.Tech., Information Science and Technology, GPA: 9.07/10 — *June 2015 - April 2019*

First class with Distinction, Rank: 10 / 120 (Top 8%); key coursework across Probability & Queueing Theory, Discrete Mathematics, Mathematics I & II, **Database Management Systems + Lab (A+ and O grades)**, Computer Architecture, Embedded Systems, Digital Principles and System Design. Independent study: Stanford CS231n (Karpathy) end-to-end with extensive PyTorch work.

Experience

Motorq — Chennai, India
Technical Lead Engineer — *January 2024 - August 2026*

- Re-architected the primary data ingestion layer to handle **600K+ msg/min**, cutting end-to-end latency from **5 min to 10 sec (30-60x)** and delivering **99.9% SLA compliance**, while reducing compute at the bottleneck and doubling vehicle-processing capacity.
- Led the **System Integration team of 7** in standing up production observability on **Victoria Metrics + Grafana**, which drove a **30% pipeline-reliability increase** and cut error rates from **40% to <0.5%** on vehicle-onboarding workflows.
- Consolidated the customer-facing distributed enrollment system from **80+ processors to 3-4** running at **10K async enrollments/min**, cutting infrastructure cost by **$10K/month**, shortening load-testing cycles from **3 days to 2 hours**, and shortening client request time by **>70%**.
- Built an **AI debugging agent + intelligent monitoring** layer (RAG over logs and code) that pulls deployment, commit, and source context from Motorq's observability stack, drafts a fix, opens a PR, and iterates until CI and human review pass, achieving an agent-drafted-fix acceptance rate of **∼40%**; an LLM-based anomaly-detection layer on top of **Victoria Metrics + Grafana** lifted detection coverage by **10%**.
- Led the design of an **Event Hub to Delta Lake** streaming architecture that replaced cron-based ETL pushing petabytes into Snowflake, saving **$5K/month** and cutting sync latency from **1 hour to <5 minutes**; my work on this architecture was selected to present at **Data Streaming Summit 2025 (StreamNative)**.

Motorq — Chennai, India
Software Engineer (I) — *December 2021 - December 2023*

- Redesigned the vehicle-onboarding workflow as a **deterministic finite automaton (DFA)** with plug-and-play states and recursive retries, which cut the error rate from **30% to 10%** and source-integration time from **5-10 days to 2-3 days**; a self-generating test suite inspired by DeepCoder drove a **50% reduction in QA + production bugs**.
- Built a **self-generating test framework** that covers **24K unique states**, which shortened the dev to deploy to validation cycle from **months to hours**; this was the first system at Motorq enabled with CI/CD.
- Built a secure ops platform that became a first-class customer of Motorq (used 24/7 by the support team) and cut processing time from **days to <5 min**; led a team of 5 UI engineers through a customer-portal React migration that improved stability by **20%**, and the portal still serves high data volumes to date.
- Shipped a **Kinesis** ingestion redesign with a heuristic state-machine filter that drops zero-contribution messages at the bottleneck, which reduced compute at the bottleneck and doubled the system's vehicle-processing capacity.

- Built a real-time map-based visualization that renders ∼**100K live vehicles** across different zoom levels with throttled position updates, server-side clustering that keeps cluster boundaries stable as vehicles move, hover-surfaced vehicle metadata, and near-real-time updates end-to-end.

Myntra Bangalore, India
Software Engineer / Senior Software Engineer *July 2019 - December 2021*

- Designed a load-testing framework, benchmarked the checkout microservice, and de-provisioned redundant servers, which together drove a **90%** reduction in server and infrastructure costs at the same load.
- Designed and shipped a client-side **StatsD**-based library for real-time backend health monitoring, which cut MTTR from **24 hours to seconds**, drove a **50%** reduction in customer escalations, and led to my promotion to Senior SWE in <2 years.
- Audited **100+ production servers** to isolate a single 'rogue' process that had silently failed to deploy months earlier and was still serving stale logic, which prevented a likely multi-million-dollar outage.
- Implemented a new payment processing pipeline that cut payment-gateway failures by **30%** and improved checkout reliability.
- Built targeted-acquisition coupons and a change-impact library that focuses QA on highest-risk changes, which together lifted checkout reliability during peak sale campaigns.

National Informatics Centre (NIC) Chennai, India
Research Intern (Routing Algorithms) *May 2017*

- Implemented a new load-balancing algorithm on the existing routing infrastructure that increased system throughput by **20%**; validated the latency and load distribution end-to-end in collaboration with the DevOps team.

Research & Technical Projects

Samsung IoT Security Research | BLE / Zigbee + Docker Emulation

- Researched BLE and Zigbee protocols independently at Samsung Research Institute, Bangalore (May - Jul 2018), identified and patched a security vulnerability in zero-touch device-setup flows, and built a Docker-based emulation platform for attack-vector testing; earned a **Pre-Placement Offer (PPO)** from Samsung after clearing the advanced competitive-programming test, making me one of few PPOs awarded in India.

DeepCoder (Neural Program Synthesis) | Best Project Award, 50+ teams

- Re-implemented Microsoft's DeepCoder paper, which models programming as a search problem over a DSL and uses AI to prune the search space, and won the **Best Project Award** at the CEG inter-department symposium among **50+ teams**; the same modeling approach directly inspired the production self-generating test framework at Motorq, and built the project in PyTorch + TensorFlow while completing Stanford CS231n (Karpathy) end-to-end during my undergrad with extensive PyTorch work.

Robotic Hand Simulation + Autonomous Robot (Edge ML) | Raspberry Pi / Jetson Nano

- Built a hand-gesture-recognition + palm-replicating robotic arm via computer-vision feature extraction on **Raspberry Pi** (which earned me a **CEG Robotics Club runner-up**), and independently built a personal-project autonomous robot with camera-based collision-avoidance on **Jetson Nano**, where I learned **ONNX** / **H5** model-weight formats and edge model deployment.

Intelligent Flood Management System (IFMS) | IoT + LIDAR

- Designed an **IoT + LIDAR** flood-risk-mitigation system inspired by the 2015 Chennai floods, and presented the design at **ISSE at SDSC SHAR (ISRO)**.

Technical Skills

Languages & Backend: TypeScript, JavaScript (Node.js production), SQL, Python (tooling), Bash; C++ (CP); Microservices, REST / gRPC, Express, React
Data, Streaming & Cloud: PostgreSQL, Snowflake, Delta Lake, ETL / ELT; AWS Kinesis, Kafka, Pulsar, Azure Event Hub, SQS; Docker, Kubernetes, AWS (EC2, S3, Lambda), Azure, Linux, CI/CD
Observability & AI/ML: Victoria Metrics, Grafana, StatsD, Proactive Alerting; PyTorch, TensorFlow, LLM Tool-Use, RAG, Agent Workflows
Embedded & Foundations: Raspberry Pi, Jetson Nano, Arduino, ONNX, H5; Data Structures, Algorithms, OOP, State-Machine / DFA Modeling

Leadership & Volunteering

- Served as **International Service Director**, Rotaract Club of CEG (2018 - 2019), where organized **Sangarsh** (raising **INR 5 Lakhs / USD 6,000+** for muscular dystrophy-affected children with **1500+ footfall**) and **Conclave** (with **1000+ footfall**), raising half a million INR total; also served as **Treasurer** of ACM CEG where organized the **Codher** women-in-tech hackathon, and as a **Core member** of CEGLUG (2017 - 2018) where mentored junior students and ran technical workshops.
- Served as a **STEM mentor** for the Prodigy / Insight program, where delivered project-based STEM workshops to K-12 students across multiple cohorts; also served as a **Curriculum committee member** for Regulations 2019 (CEG undergrad, faculty-selected), where reviewed and updated the undergraduate syllabus.

Honors & Recognition

- Secured a **Pre-Placement Offer (PPO)** from Samsung Research Institute, Bangalore after clearing the advanced competitive-programming test; this PPO was one of few awarded in India.
- Won the **Best Project Award** for DeepCoder (neural program synthesis) at the CEG inter-department symposium, earning Best Project among **50+ teams**.
- Earned Second runner-up (team) at the **Chargebee Hackathon** - Women Tech Fab; team was the only non-professional team in the hackathon.
- Reached Finalist (team) at the **Accenture Blockchain Hackathon**.
- Reached Finalist (internal hackathon) at **Myntra HackerRamp**.
- Selected as a **Speaker** at the Data Streaming Summit 2025 (StreamNative), where presented the petabyte-scale Event Hub $\rightarrow$ Delta Lake streaming architecture built at Motorq.